



DSA 104 AI and ML in Chemistry

Session 24: Recap 5

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```
import tensorflow as tf

model.add(Dense(64, activation='relu'))

optimizer = tf.keras.optimizers.Adam(),

model.compile(loss='categorical_crossentropy',

model.fit(X_train, y_train, epochs=10)
```

Points for today

Discussion Exercise 10

Discussion Mock exam

Symposium

Outlook

Feedback and course discussion



Universität
Zürich^{UZH}

Department of Chemistry



DSA Symposium: Automation and Data Science in Action

Lecture room Y11-F-06

27 May 2026, 08:30 – 16:00

DSA Symposium – Agenda:

27 May 2026
Y11-F-06

- 08:30 – 09:15 *Advancing hair and nail matrix analysis with automated workflow solutions*
Tina Binz (UZH)
- 09:15 – 10:00 *Biocatalysis as a Tool in Synthesis: Continuous-Flow Routes to Greener Chemistry*
Glenn Bojanov (University of Bern)
- 10:00 – 10:30 *Coffee break*
- 10:30 – 11:45 *Tracking plant volatiles in real time: New tools for studying biotic interactions*
Tristan Cofer (University of Bern)
- 11:45 – 12:00 *Optimising Chemical Process Development with Automation and Machine Learning: An Industrial Perspective*
Joshua W. Sin (F. Hoffmann-La Roche Ltd)
- 12:00 – 13:00 *Lunch for speakers and registered participants*
- 13:00 – 13:40 *Machine learning for predictive ecotoxicology*
Christoph Schür (Eawag)
- 13:40 – 14:20 *Generative machine learning for catalyst discovery (virtual lecture)*
Sarina Kopf (EPFL)
- 14:20 – 15:00 *From automation to autonomy: the rise of Self-Driving Labs®*
Loïc Roch (Atinary Technologies Inc.)
- 15:00 – 16:00 *Apéro (Y38-G-32) and lab tours*

Taste for more? Get involved!

Basic knowledge of all the tools for your own data science...

- Now: Ready to deploy the knowledge! (Thesis!)
- Ready for a new challenge? **SDL Hackathon!!** (See last slide)

More courses of the DSA Minor – of course not only for chemists:

- DSA 102: Lab Automation and Accelerated Chemistry
- DSA 103: Advanced Chemical Data Science (4 ECTS, fall semester)
- DSA 105: Automation and Chemical Data Science in practice (4 ECTS)

Exciting Electives:

- **NEW: DSA 301 Automating Chemistry – HTEL Live! (4 ECTS, DSA 102 needed)**
- CHE 725 Hacking for Chemists: Prototyping and Hardware Programming (2 ECTS, fall semester)
- **NEW: DSA 302 Hacking the HTEL – Prototyping in Action! (2 ECTS, CHE 725 highly recommended)**



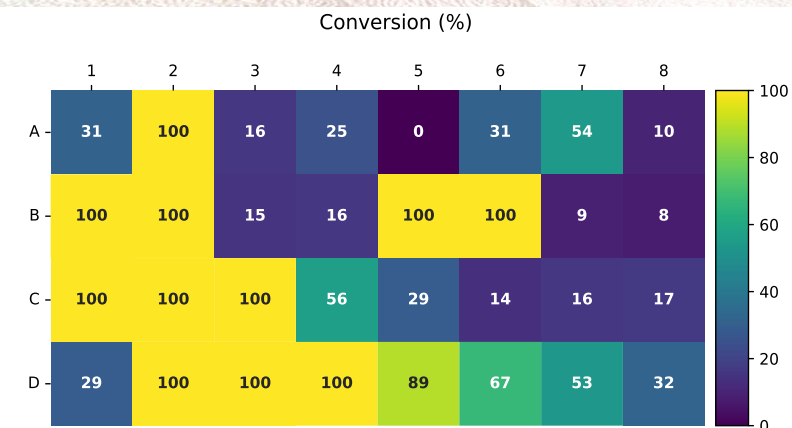
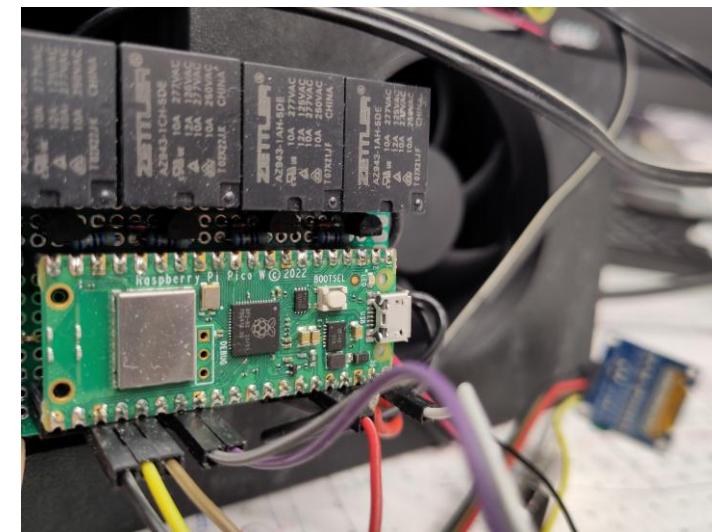
Looking for a thesis? Join HTEL and be part of the revolution!

Why?

- Lab meets data science – high demand!!
- State-of-the-art infrastructure
- High-value collaborations with AI developers and digital chemistry groups
- Multidisciplinary research
- 3D prototyping and custom solutions
- **We have robots** 😊

What?

- Printing catalysts: Metal-ionic liquids
- Catalytic LEGO: Building-block catalysis
- Enriching big reaction data for AI agent development, e.g. Suzuki couplings
- Grow beautiful crystals: Development of HT crystallisation
- Getting the analysis down: Efficient HTE quantification
- Technical projects: low level robotics, custom integrations (OT2), ML applications, ...



Self-driving labs Hackathon by EPFL LIAC and UZH HTEL

Interested in a new challenge?

Join the world's first practical lab automation hackathon!

How to participate:

- 1) Have a look at the [announcement](#)
- 2) Gather some teammates
- 3) Think of some cool research projects
- 4) Apply (same link as above) – deadline May 31
- 5) Just do it 😊
- 6) Win a prize?

Feedback and course discussion

Feedback?

Course format?

Guest contributions?

Take aways?